

Vector Integration: Trigonometric Parameterization and Wallis' Formula Shortcuts

1. Geometric Parameterization and Walli's Reduction Formulas

Name of Concepts

1. Parametric Coordinate Transformations (Circle vs. Ellipse).
2. Symmetric Definite Integral Multipliers over a Full Period (2π).
3. Wallis' Reduction Formulas for Even Powers of Sines and Cosines.

Core Mathematical Formulas

For a circle centered at the origin with radius a :

$$x^2 + y^2 = a^2 \implies x = a \cos \theta, \quad y = a \sin \theta$$

For an ellipse centered at the origin with semi-major axis a and semi-minor axis b :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \implies x = a \cos \theta, \quad y = b \sin \theta$$

The symmetry transformation rule for an even, periodic function integrated over a full circle loop is:

$$\int_0^{2\pi} \sin^n \theta \, d\theta = 4 \int_0^{\pi/2} \sin^n \theta \, d\theta \quad (\text{where } n \text{ is an even integer})$$

Wallis' shortcut template for evaluating even powers integrated across the first quadrant boundary $[0, \frac{\pi}{2}]$ is:

$$\int_0^{\pi/2} \cos^n \theta \, d\theta = \frac{(n-1) \cdot (n-3) \dots 1}{n \cdot (n-2) \dots 2} \times \frac{\pi}{2}$$

Beginner-Friendly Explanation of Operations

1. **Parameterization Refresher:** When evaluating boundary paths for Green's, Stokes', or standard line integrals, we convert Cartesian coordinates (x, y) into matching angular coordinates (θ) using standard geometric forms. Circles scale evenly using a single constant radius (a), while ellipses use two distinct axial stretch values (a and b). Tracing a complete closed loop always means running the angle parameter from $\theta = 0$ directly to $\theta = 2\pi$.
2. **Exploiting Quadrant Symmetry (The "Factor of 4" Trick):** If you sketch out the curves for $\sin^2 \theta$, $\cos^2 \theta$, or any even power like $\cos^4 \theta$, you will notice that the area bounded underneath the wave shape repeats identically in each of the four quadrants. Instead of integrating across the entire clumsy range from 0 to 2π , we can evaluate the area of just the very first clean quadrant from 0 to $\frac{\pi}{2}$, and then multiply the result by 4.
3. **Wallis' Formula Shortcut:** Integrating expressions like $\cos^4 \theta$ by hand requires using long, messy double-angle expansion rules over and over.

Wallis' reduction formula bypasses this algebra entirely with a simple fraction pattern:

- **Numerator:** Start with your exponent minus one ($n - 1$), then count down by subtracting two each time until you hit 1.
 - **Denominator:** Start with the raw exponent (n), then count down by subtracting two each time until you hit 2.
 - **The Scaling Factor:** If your original exponent is even, multiply the entire fraction chain by $\frac{\pi}{2}$ at the end.
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2. Step-by-Step Evaluation of Definite Integrals

Example A: Integrating $\sin^2 \theta$ over a full loop

Step 1: Apply the quadrant symmetry rule

Convert the full boundary sweep into an isolated first-quadrant integration multiplied by 4:

$$\int_0^{2\pi} \sin^2 \theta \, d\theta = 4 \int_0^{\pi/2} \sin^2 \theta \, d\theta$$

Step 2: Set up Wallis' formula parameters

Identify the power value: $n = 2$.

- The numerator product sequence stops at: $n - 1 = 2 - 1 = 1$.
- The denominator product sequence stops at: $n = 2$.

Step 3: Compute the fractional products

$$\int_0^{\pi/2} \sin^2 \theta \, d\theta = \frac{1}{2} \times \frac{\pi}{2}$$

Step 4: Scale by the symmetry multiplier to find the final result

Multiply the calculated single quadrant value by our 4-quadrant factor:

$$\begin{aligned} \int_0^{2\pi} \sin^2 \theta \, d\theta &= 4 \times \left[\frac{1}{2} \times \frac{\pi}{2} \right] \\ \int_0^{2\pi} \sin^2 \theta \, d\theta &= 4 \times \left[\frac{\pi}{4} \right] = \pi \end{aligned}$$

Example B: Integrating $\cos^4 \theta$ over a full loop

Step 1: Apply the quadrant symmetry rule

Convert the full loop range into a single quadrant calculation scaled by 4:

$$\int_0^{2\pi} \cos^4 \theta \, d\theta = 4 \int_0^{\pi/2} \cos^4 \theta \, d\theta$$

Step 2: Set up Wallis' formula parameters

Identify the power value: $n = 4$.

- Build the numerator by starting at $(4 - 1) = 3$ and subtracting two: $3 \cdot 1$.
- Build the denominator by starting at 4 and subtracting two: $4 \cdot 2$.

Step 3: Compute the fractional products

Combine the descending chains and multiply by the even power scaling constant $\frac{\pi}{2}$:

$$\int_0^{\pi/2} \cos^4 \theta \, d\theta = \left[\frac{3 \cdot 1}{4 \cdot 2} \right] \times \frac{\pi}{2} = \frac{3}{8} \times \frac{\pi}{2} = \frac{3\pi}{16}$$

Step 4: Scale by the symmetry multiplier to find the final result

Multiply the calculated single quadrant value by our 4-quadrant factor:

$$\int_0^{2\pi} \cos^4 \theta \, d\theta = 4 \times \left[\frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \right]$$

$$\int_0^{2\pi} \cos^4 \theta \, d\theta = 4 \times \left[\frac{3\pi}{16} \right]$$

Simplify the fraction coefficients to finish the evaluation:

$$\int_0^{2\pi} \cos^4 \theta \, d\theta = \frac{3\pi}{4}$$